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BATCH-15

Question 1

Write a C program to:

- a) Read the number of rows and columns from the user. Ensure that both values are positive. If not, repeat to get the desired input(s).
- b) Declare a variable-length 2D integer array using the given dimensions.
- c) Read the array elements column by column using nested loops, with the outer loop iterating over columns and the inner loop over rows.
- d) Print the elements in row-major order using a single loop and a linear index.

AIM

To write a C program to read a variable-length 2D array column-wise and display its elements in row-major order using a single loop.

Algorithm

1. Start.
2. Read the number of rows and columns.
3. If rows or columns are not positive, ask the user to enter them again.
4. Declare a variable-length 2D integer array.
5. Read the array elements column by column using nested loops.
6. Display the elements in row-major order using a single loop and linear index.
7. Stop.

>C program:

```
#include <stdio.h>

int main()
{
    int rows, cols;
    do
    {
        printf("Enter number of rows and columns: ");
        scanf("%d %d", &rows, &cols);
    } while(rows <= 0 || cols <= 0);
```

```

int a[rows][cols];

printf("Enter array elements column-wise:\n");

for(int j = 0; j < cols; j++)

{

    for(int i = 0; i < rows; i++)

    {

        scanf("%d", &a[i][j]);

    }

}

printf("Elements in row-major order:\n");

for(int k = 0; k < rows * cols; k++)

{

    printf("%d ", a[k / cols][k % cols]);

}

return 0;

}

```

>SAMPLE EXECUTION:

Enter number of rows and columns: 2 3

Enter array elements column-wise:

1

2

3

4

5

6

Elements in row-major order:

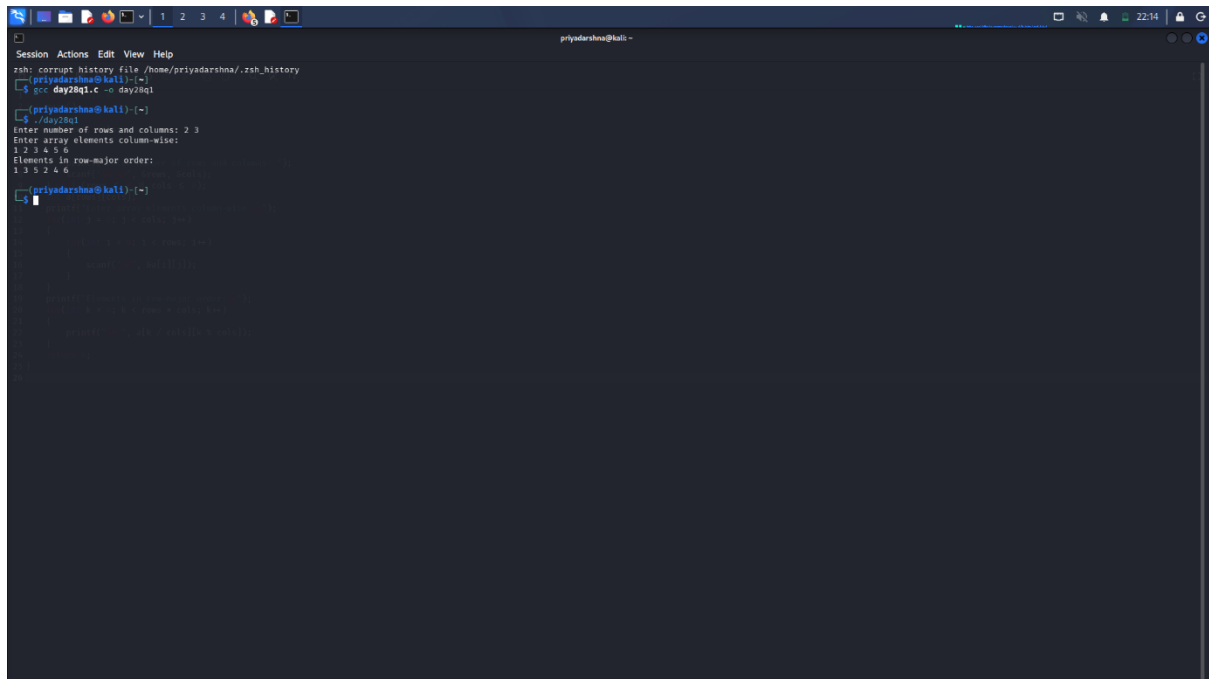
1 3 5 2 4 6

>TEST CASE TABLE:

Test Case	Rows	Columns	Input Elements (column-wise)	Output (row-major)
1	2	3	1 2 3 4 5 6	1 3 5 2 4 6

Test Case	Rows	Columns	Input Elements (column-wise)	Output (row-major)
2	2	2	10 20 30 40	10 30 20 40
3	1	3	5 6 7	5 6 7

>OUTPUT :



```

Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day28q1.c -o day28q1
[priyadarshna@kali]~$ ./day28q1
Enter number of rows and columns: 2 3
Enter array elements column-wise:
1 2 3 4 5 6
Elements in row-major order:
1 3 5 2 4 6
[priyadarshna@kali]~$

```

Question 2

Write a C program to perform matrix multiplication of two matrices.

The program should:

- Read the dimensions and elements of two matrices.
- Check whether matrix multiplication is possible.
- Multiply the two matrices using nested loops.
- Display the resultant matrix.

Aim

To write a C program to multiply two matrices and display the resultant matrix.

Algorithm

1. Start.
2. Read the number of rows and columns of both matrices.
3. Check whether the number of columns of the first matrix is equal to the number of rows of the second matrix.
4. Read the elements of both matrices.
5. Multiply the matrices using nested loops.
6. Store the result in a third matrix.
7. Display the resultant matrix.
8. Stop.

>C program:

```
#include <stdio.h>

int main(){
    int r1,c1,r2,c2;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d",&r1,&c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d",&r2,&c2);

    if(c1!=r2){
        printf("Matrix multiplication is not possible.");
        return 0;
    }

    int a[r1][c1],b[r2][c2],c[r1][c2];

    printf("Enter elements of first matrix:\n");
    for(int i=0;i<r1;i++)
        for(int j=0;j<c1;j++)
            scanf("%d",&a[i][j]);

    printf("Enter elements of second matrix:\n");
    for(int i=0;i<r2;i++)
        for(int j=0;j<c2;j++)
```

```

        scanf("%d",&b[i][j]);
for(int i=0;i<r1;i++) {
    for(int j=0;j<c2;j++){
        c[i][j]=0;
        for(int k=0;k<c1;k++){
            c[i][j]+=a[i][k]*b[k][j];}
    }
    printf("Resultant matrix:\n");
    for(int i=0;i<r1;i++){
        for(int j=0;j<c2;j++){
            printf("%d ",c[i][j]);
            printf("\n");}
    }
    return 0;
}

```

>SAMPLE EXECUTION:

Enter rows and columns of first matrix: 2 2

Enter rows and columns of second matrix: 2 2

Enter elements of first matrix:

1 2

3 4

Enter elements of second matrix:

5 6

7 8

Resultant matrix:

19 22

43 50

>TEST CASE TABLE:

Test Case	Matrix A	Matrix B	Result
1	1 2 / 3 4	5 6 / 7 8	19 22 / 43 50
2	1 0 / 0 1	2 3 / 4 5	2 3 / 4 5
3	1 2 3	4 / 5 / 6	32

>OUTPUT:

```

Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali:~$ gcc day2bq2.c -o day2bq2
priyadarshna@kali:~$ ./day2bq2
Enter rows and columns of first matrix: 2 3
Enter rows and columns of second matrix: 3 2
Enter elements of first matrix:
1 2 3
4 5 6
Enter elements of second matrix:
7 8
9 10
11 12
Resultant matrix:
38 64
139 154
priyadarshna@kali:~$

```